# MONODROMY VIA TYPE TRANSPORT: A CUBICAL AGDA FORMALIZATION

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Abstract. We give a source-indexed Cubical Agda formalization of the constructions in a frozen version of the nLab section "Monodromy: In homotopy type theory." Starting from a family $P : B \to \mathsf{Type}$, the checked development identifies the homotopy fiber of the dependent projection, constructs transport and its equivalence structure, and records its unit and composition laws. Restriction to loops yields fiber monodromy. For a chosen fiber point, this produces a pointed equivalence to the transported basepoint and hence group equivalences of based homotopy groups. An explicit path between basepoints gives the corresponding change-of-basepoint map. No fundamental-group action or higher-coherent action structure is claimed.

## 1. Introduction

This article accompanies the verified module `NLabMonodromy.agda`. Its scope is deliberately fixed by the frozen nLab source recorded in the experiment directory [1]. The source was available while this paper was generated; hence this is a *source-informed reconstruction*, not a blind reverse-translation evaluation. The purpose is to exhibit, declaration by declaration, what the checked code expresses and what it does not express.

We use the standard perspective in which a dependent family is a fibration in the homotopy-theoretic sense of type theory [2, §2.3]. This does not assert the classical Serre lifting property. The relevant equality structure is supplied by the type theory and made explicit below.

## 2. Type families and homotopy fibers

Let $B : \mathsf{Type}$ and $P : B \to \mathsf{Type}$. Set

$$E := \sum_{b:B} P(b), \qquad \mathsf{pr}_1 : E \to B. \tag{1}$$

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**Preliminary Draft — Mathematical and Semantic Review Pending.** AI assistance was used in preparing the formalization and this source-informed reconstruction. It is not a mathematical author, and this attribution is not evidence of completed human review.

For $b : B$, the dependent projection occurs in the homotopy-pullback square

$$\begin{array}{ccc} P(b) & \longrightarrow & E \\ \downarrow & & \downarrow{\scriptstyle \mathsf{pr}_1} \\ * & \xrightarrow[b]{} & B. \end{array}$$

The formal evidence is not a strict pullback assertion, but the checked equivalence

$$\mathrm{hofib}_{\mathsf{pr}_1}(b) \simeq P(b), \qquad \mathrm{hofib}_f(b) :\equiv \sum_{e:E}(f(e) = b). \tag{2}$$

In the module, the source-indexed declaration is `total-hofib≃`. It is obtained from the Cubical Library declaration `fiberProjEquiv`, whose orientation is reversed by the library operation `invEquiv`. The local projection is exactly the following checked code excerpt.

```
total-projection P = fst
```

## 3. Transport and monodromy

For $p : x =_B y$, transport in $P$ gives a function

$$p_* : P(x) \longrightarrow P(y). \tag{3}$$

The local declaration `transport-map` is intentionally kept distinct from `transport-equiv`, the latter being the equivalence $P(x) \simeq P(y)$. The Cubical Library supplies the equivalence package through `substEquiv`; its inverse data is mathematically transport along the inverse path. The actual local definition of the function is

```
transport-map P p = subst P p
```

The two elementary compatibilities are checked as pointwise paths. Thus for $u : P(x)$,

$$(\mathsf{refl}_x)_*(u) = u, \tag{4}$$

and, for $p : x = y$ and $q : y = z$,

$$(p \cdot q)_*(u) = q_*(p_*(u)). \tag{5}$$

Here Cubical Agda's $p \cdot q$ means first $p$, then $q$. Equation (5) is represented by `transport-comp`, using the library path `substComposite`; it is the evidence for the triangle

$$\begin{array}{ccc} P(x) & \xrightarrow{p_*} & P(y) \\ & {\scriptstyle (p\cdot q)_*} \searrow & \downarrow{\scriptstyle q_*} \\ & & P(z). \end{array}$$

These are propositional, pointwise equalities. They are not claims that the two maps are definitionally equal.

Fixing $b_0 : B$ and writing $F = P(b_0)$, transport along loops defines

$$\mathrm{mon}_{P,b_0} : \Omega(B, b_0) \longrightarrow \mathrm{Aut}(F), \qquad \gamma \longmapsto \gamma_*. \tag{6}$$

The declaration `monodromy-map` uses the library pointed loop space `Omega`; `monodromy-refl` and `monodromy-comp` provide the corresponding pointwise unit and composition paths. No higher coherence data and no descent from the loop space to $\pi_1(B, b_0)$ is present.

## 4. Pointed transport and homotopy groups

Let $f_0 : F$ and $gamma : b_0 = b_0$. Put $f_1 \coloneqq \gamma_*(f_0)$. The important point is that $f_1 = f_0$ is not a consequence of transport. The checked statement is instead the pointed equivalence

$$(F, f_0) \simeq_* (F, f_1), \tag{7}$$

represented by `pointed-monodromy`. Its basepoint witness is definitionally the reflexivity path because the target is based at the transported point.

The library construction `piIso` now yields a *group* equivalence, not merely an equivalence of carriers:

$$\pi_n(F, f_0) \cong \pi_n(F, f_1), \qquad n \geq 1. \tag{8}$$

The local declaration is `pointed-monodromy-π`. Cubical Library's `piGr n` presents conventional $\pi_{n+1}$, so the natural-number index used in the code covers exactly the range $n \geq 1$ in (8).

Finally, if a path $r : f_1 = f_0$ is supplied, `basepoint-change•` and `basepoint-change-π` give the corresponding pointed equivalence and group isomorphism. The hypothesis $r$ remains explicit. In particular, these conditional maps do not supply an unconditional fundamental-group action.

## 5. Formalization and verification

The formal module has ten source statement identifiers, `MON-HOTT-001` through `MON-HOTT-010`. The total-space, transport, and monodromy declarations are local source-indexed wrappers; the fiber, transport, pointed-type, and homotopy-group mechanisms are attributed Cubical Library results. The pinned environment is Agda `2.8.0-3d04bac` and Cubical Agda Library v0.9, commit `87af036ebd1c87956cf08fea44e62c7cf76392ca`.

The module type-checks with the command recorded in the experiment report. It has no holes, unsolved metas, local postulates, or termination overrides. The JSONL alignment records give full Agda types, source locations, assumptions, dependencies, and translation classifications for every frozen claim.

## 6. Limitations

Machine checking establishes that the formal declarations are accepted in the pinned imported context. It does not establish that this exposition and the frozen source are semantically identical. Mathematical correctness, formalization correctness, and semantic alignment remain pending review by Hongwei (Alex) Wang (REV-001).

The present code does not produce a fundamental-group action, show representative-independence after truncation, choose basepoint-return paths, or verify their compatibility with loop composition. Those omissions are intentional boundaries of the checked result, not suppressed proof gaps.

## References

[1] nLab authors, *Monodromy*, "In homotopy type theory." https://ncatlab.org/nlab/show/monodromy#InHomotopyTypeTheory.

[2] The Univalent Foundations Program, *Homotopy Type Theory: Univalent Foundations of Mathematics*, Institute for Advanced Study, 2013. https://homotopytypetheory.org/book/.

[3] Cubical Agda Library contributors, *Cubical Agda Library*, version 0.9, commit 87af036ebd1c87956cf08fea44e62c7cf76392ca.

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